How the UV Sphere Is Built

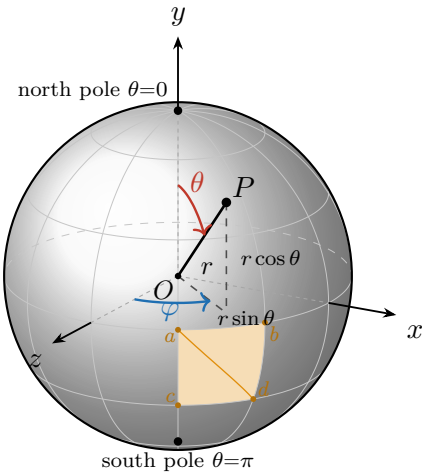


Figure 1. Any point P on the sphere is picked by two angles: θ (down from the north pole) and φ (around the y axis, starting at $+z$). The orange patch is one grid cell.

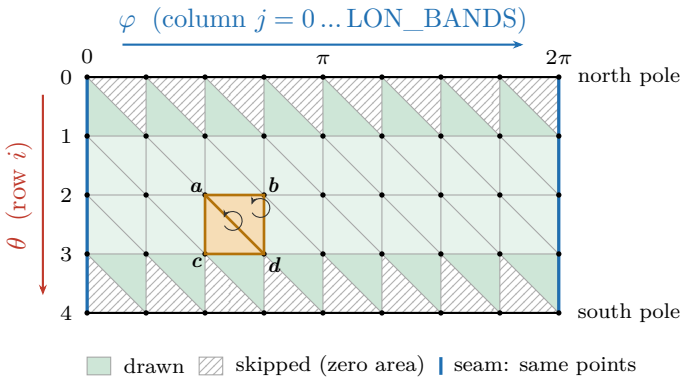


Figure 2. The sphere "unwrapped" into its grid of angles (drawn with only 4×8 cells). Every cell becomes two triangles. In the top and bottom rows one triangle has zero area and is skipped.

1. Every point comes from two angles

Instead of listing points by hand, we describe each point by *where it is*: how far down from the top (θ, from 0 at the north pole to π at the south pole) and how far around (φ, from 0 to 2π). With y pointing up:

$$x = r \sin\theta \, \sin\varphi, \qquad y = r\cos\theta, \qquad z = r \sin\theta \, \cos\varphi$$

Read it off Figure 1: P sits at height $r\cos\theta$, and at distance $r\sin\theta$ from the y axis. That distance is then split between x and z by the angle φ.

2. Walk a grid of angles

Split θ into `LAT_BANDS` equal steps and φ into `LON_BANDS` equal steps. Each pair (i, j) makes one vertex. The vertices are stored row by row, so vertex (i, j) is at

$$\text{index}(i, j) = i \cdot (\text{LON_BANDS} + 1) + j.$$

The last column ($\varphi = 2\pi$) lands on top of the first ($\varphi = 0$). We store it twice anyway, because that keeps the index math simple. With 24×48 bands that is $25 \times 49 = 1225$ vertices.

3. Two triangles per cell

The four corners of a cell are $a = (i, j)$, $b = (i, j{+}1)$, $c = (i{+}1, j)$, and $d = (i{+}1, j{+}1)$. The cell (orange in both figures) is cut along the diagonal a–d into triangles

$$(a,\ c,\ d) \quad \text{and} \quad (a,\ d,\ b).$$

Seen from *outside* the sphere, both triangles go **counter-clockwise**. That makes them front faces, so `cullMode: 'back'` throws away only the triangles on the far side.

4. Take care of the poles

Every vertex in row $i = 0$ is the north pole, so there a and b are the same point. The triangle (a, d, b) would be a flat line with zero area, so we skip it. The same thing happens at the south pole ($c = d$), where we skip (a, c, d). Total: $2 \cdot \text{LON_BANDS} \cdot (\text{LAT_BANDS} - 1) = 2208$ triangles.

5. Normals (and colors) for free

For a sphere centered at the origin, the outward normal is just the direction from the center to the point:

$$\mathbf{n} = \frac{P}{r} = (\sin\theta\sin\varphi,\ \cos\theta,\ \sin\theta\cos\varphi).$$

We don't need cross products. Neighboring triangles *share* vertices and normals, so the fragment shader blends the normals smoothly across each triangle. That is why the sphere looks round even though it is made of flat triangles. (The tetrahedron is different: each face has its own copies of its vertices, with one flat normal.) The color is the normal moved from $[-1, 1]$ into $[0, 1]$: $\text{color} = 0.5\,\mathbf{n} + 0.5$.

Each vertex is 9 floats: `X Y Z | NX NY NZ | R G B`.